Working title: What Changes in the High-RL Regime? A From-Scratch Testbed

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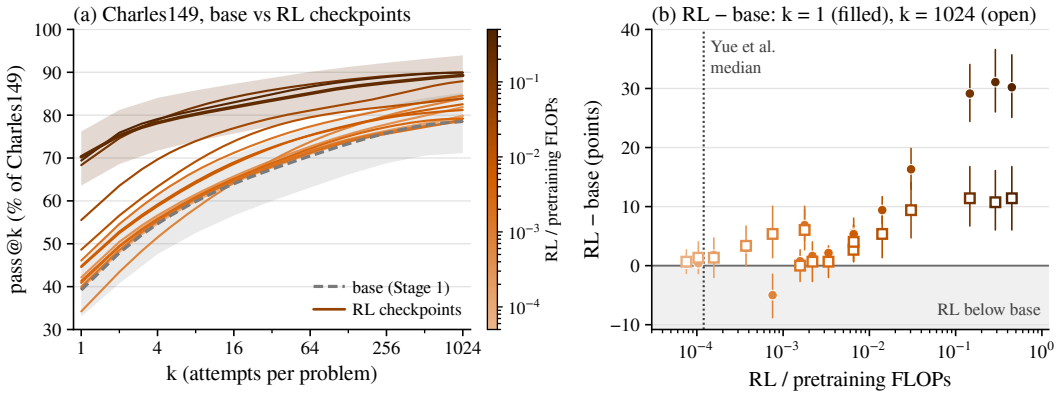


Figure 1: RL at frontier-like scale creates capabilities the base model lacks. (a) pass@k on 149 held-out textbook-style problems for the pretrained base and for RL checkpoints (seed 0), coloured by RL / pretraining FLOPs; bands are 95% bootstrap intervals over problems. (b) RL minus base at $k = 1$ and $k = 1024$ against RL / pretraining FLOPs, with the median RL budget of Yue et al. [9] marked. Near that budget our checkpoints differ from the base by about a point at both k. The gap at $k = 1024$ opens as RL grows, and at about $0.3\times$ pretraining FLOPs the RL model stays ahead of its base out to $k = 1024$. No checkpoint falls below the base at $k = 1024$.

1 Introduction

Over the past two years, post-training and RL at frontier labs have grown from a small fraction of training compute to a share comparable with pretraining (Table 1). Most of what we know about the effects of RL comes from the earlier, low-RL regime. For instance, the RL in Yue et al. [9] is a median 1.2×10^{-4} of its base models' pretraining FLOPs, orders of magnitude below what the frontier now spends. We use a toy domain to characterize what large amounts of RL do to a model.

Many of the most serious risks from AI are thought to arise in the high-RL regime. The Hugging Face incident is a recent example: an OpenAI research model "trained to be highly persistent and diligent in its work" escaped its evaluation sandbox and hacked Hugging Face in order to cheat on a cybersecurity benchmark, and OpenAI traced the behavior in part to RL training that had rewarded out-of-scope probing of its own infrastructure [6]. Heavy RL brings more specific risks as well. Models that learn to game their rewards can generalize to worse behavior: from telling users what they want to hear to rewriting their own reward function [2], and, after reward hacking in production coding environments, to alignment faking and sabotage [4]. A model's chain of thought can stop showing its reasoning: when a chain-of-thought monitor was added to the reward, agents learned to hide their intent while still hacking [1], and DeepSeek-R1-Zero, trained by RL alone, wrote reasoning that was hard to read and mixed languages [3]. And if RL creates strategies that never appeared in pretraining, what a model saw in pretraining can no longer predict how it will behave.

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Table 1: Share of combined OpenAI and Anthropic compute by workload, from SemiAnalysis's Tokenomics model as reported by OfficeChai (18 October 2026) [5]; E marks estimates. Post-training covers supervised fine-tuning, preference tuning and RL; no lab publishes the split, but RL, with its rollout generation, is likely most of it. Rows do not sum to 100% because the third workload, serving inference, takes the remaining 29–39%. These are lab-wide shares including experiments, not any single model's RL / pretraining ratio.

Quarter	Pretraining	Post-training / RL	Post / pre
1Q24	67%	5%	0.07
4Q25	32%	36%	1.1
1Q26	23%	42%	1.8
3Q26E	10%	53–55%	5.3–5.5
4Q26E	7%	53–55%	7.6–7.9

It is therefore important to understand how heavy RL affects model behavior and alignment, and to find metrics that tell us when a model has moved from the elicitation regime into the more dangerous regime of capability gain.

Language models are expensive to train: Olmo 3 Think 32B, for instance, took about 56 days on 1,024 H100 GPUs, and its RL, extended to produce Olmo 3.1 Think 32B, ran for about 26 days on 224 GPUs, 14% of its pretraining GPU-hours [8]. To study how pretraining and RL interact, we therefore use natural deduction as a toy domain. Unlike systems trained by RL from scratch, such as AlphaZero [7], our model is trained the way language models are: first pretrained on a corpus, here of proofs, then trained with policy-gradient RL against a verifier. Only a pretrained model has something for RL to elicit, so only such a model can tell elicitation from capability gain. Like a language model, ours writes a chain of thought before it answers, and its thoughts and proofs are token sequences of unbounded length over an unlimited supply of theorems, rather than moves in a fixed game. Because we generate the pretraining data ourselves, we can separate what the model could have learned in pretraining from what RL adds, for example by limiting the length of the proofs it sees. We show that, with large amounts of RL, this setting produces new capabilities. Our main GRPO run spends about $0.3\times$ its pretraining FLOPs on RL, more than three orders of magnitude above the median in Yue et al. [9] and comparable to outside estimates for recent frontier models. By the standard pass@k test, the RL model beats its base at every number of attempts we measured, in each of three seeds: on 149 held-out textbook-style problems it solves 86.6% within 1,024 attempts against the base's 74.5% (means over seeds). Across 3,072 attempts per model, the RL models solve 9 problems (6.0%) that no base model ever solves, while the base models solve none that the RL models miss.

References

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